

metabolite_cluster_super_pathway copy

Cluster	Classes	Metabolite	
1	Amino Acids	2-hydroxy-3-methylvalerate	3-methyl-2-oxobutyrate
		3-methyl-2-oxovalerate	4-methyl-2-oxopentanoate
		alpha-hydroxyisocaproate	alpha-ketobutyrate
		n-acetylglycine	
	Lipids	3-hydroxybutyrate (bhba)	3-hydroxybutyrylcarnitine (1)
		3-hydroxybutyrylcarnitine (2)	acetoacetate
		acetylcarnitine	c2 (acetylcarnitine)
	Nucleotides	3-aminoisobutyrate	
	Xenobiotics	2,3-dihydroxyisovalerate	
2	Amino Acids	3-(4-hydroxyphenyl)lactate	3-phenylpropionate (hydrocinnamate)
		indoleacetylglutamine	indolepropionate
		ornithine	phenyllactate (pla)
		pro-hydroxy-pro	trans-urocanate
		urea	xanthurenate
	Cofactors and Vitamins	trigonelline (n'-methylnicotinate)	
	Lipids	4-androsten-3alpha,17alpha-diol monosulfate (2)	4-androsten-3beta,17beta-diol monosulfate (1)
		choline	cortisol
		lysopc a c14:0	lysopc a c16:1
		lysopc a c20:4	pregnenolone sulfate
	Peptides	gamma-glutamylglycine	gamma-glutamyltryptophan
	Xenobiotics	3-hydroxyhippurate	3-hydroxypyridine sulfate
		hippurate	methyl-4-hydroxybenzoate sulfate
		piperine	
3	Energy	phosphate	
	Lipids	1-docosaehaenoyl-gpe (22:6)	10-undecenoate (11:1n1)
		13-hode + 9-hode	15-methylpalmitate
		16-hydroxypalmitate	17-methylstearate
		3-hydroxydecanoate	3-hydroxylaurate
		3-hydroxymyristate	5-dodecenoate (12:1n7)
		caprate (10:0)	glycerol
		laurate (12:0)	
	Nucleotides	xanthine	
4	Amino Acids	2-hydroxybutyrate/2-hydroxyisobutyrate	3-indoxyl sulfate
		alanine	dopamine sulfate (2)
		isobutyrylcarnitine	n-acetylkynurenine (2)
		n-acetyltryptophan	phenylacetylcarnitine
		proline	
	Carbohydrates	pyruvate	
	Energy	malate	
	Lipids	3-hydroxy-3-methylglutarate	c0 (carnitine)
		c3 (propionylcarnitine)	c4 (butyrylcarnitine)
		deoxycholate	methylmalonate (mma)
		propionylcarnitine	
	Xenobiotics	4-hydroxyhippurate	o-methylcatechol sulfate
5	Amino Acids	n-acetylglutamate	
	Energy	cis-aconitate	
	Lipids	1-linoleoyl-gpe (18:2)	3-hydroxyoctanoate

	lysopc a c18:2	pc aa c36:1
	pc aa c36:2	
6 Lipids	adipoylcarnitine	c12 (dodecanoylcarnitine)
	c14 (tetradecanoylcarnitine)	c14:1 (tetradecenoylcarnitine)
	c14:1-oh (hydroxytetradecenoylcarnitine)	c14:2 (tetradecadienylcarnitine)
	c16:2 (hexadecadienylcarnitine)	c18:1 (octadecenoylcarnitine)
	dodecanedioate	hexadecanedioate
	myristoleoylcarnitine	myristoylcarnitine
	octadecanedioate	oleoylcarnitine
	palmitoylcarnitine	sebacate (decanedioate)
	suberoyl carnitine	tetradecanedioate